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Oefeningenreeks 4A nr. 4.10

$$\begin{aligned} & \int \frac{x^3 \sqrt[5]{x^3}}{\sqrt[3]{x}} \\ &= \int \frac{x^3 \cdot x^{\frac{3}{5}}}{x^{\frac{1}{3}}} dx = \frac{x^{\frac{18}{5}}}{x^{\frac{1}{3}}} dx = \int x^{\frac{18}{5} - \frac{1}{3}} dx = \int x^{\frac{49}{15}} dx \\ &= \frac{15}{64} x^{\frac{64}{15}} + C = \frac{15}{64} \cdot x^{\frac{60}{15}} x^{\frac{4}{15}} + C = \frac{15}{64} x^4 \sqrt[15]{x^4} + C \end{aligned}$$

References